Real Estate Data Analysis

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1. Introduction

- *Overview of the project and its objective in analyzing the USA Real Estate dataset to extract insights and support business decision-making.*

2. Dataset Description

- *Explanation of the dataset source, its purpose, and detailed description of all columns used in the analysis.*

3. Data Warehouse Overview

- *Brief introduction about the Data Warehouse concept and how it supports analytical processing (OLAP) for decision making.*

4. Tools Used

- *Overview of the tools used in the project:*
  - *Alteryx*
  - *SSIS (SQL Server Integration Services)*

5. Data Splitting Process

- *Explanation of how the dataset was divided into two halves using Alteryx for parallel processing between Alteryx and SSIS.*

6. Data Processing in Alteryx

- *Detailed explanation of:*
- *Data cleaning*
- *Data transformation*
- *Feature engineering*
- *Filtering and outlier handling*
- *Aggregations and summarization*

- *Output to SQL Server*

7. Data Processing in SSIS

- *Explanation of SSIS workflow including:*
  - *Data input*
  - *Derived column transformations*
  - *Data conversion*
  - *Conditional splitting*
  - *Loading data into SQL Server*

8. Data Transformation Techniques

- *Description of applied transformations such as:*
  - *Handling missing values*
  - *Formatting text fields*
  - *Splitting date attributes*
  - *Filtering invalid data*

9. Data Analysis & Insights

- *Business insights extracted from the dataset such as:*
  - *Pricing patterns*
  - *Property characteristics impact*
  - *Location-based trends*
  - *Market behavior analysis*

10. Decision Making

- *Key business decisions derived from charts and analysis results.*

11. Conclusion

- *Summary of the project outcome and value of using data warehouse tools for real estate analytics.*

1. Dataset Description

- The dataset used in this project is the USA Real Estate Dataset obtained from Kaggle(https://www.kaggle.com/datasets/ahmedshahriarsakib/usa-real-estate-dataset). It contains approximately 2.5 million records describing real estate property transactions across different states in the United States.
- This dataset is used to analyze housing market trends, identify pricing patterns, and support business decision-making in the real estate domain.

Columns Description:

- **brokered_by**: ID of the broker responsible for the transaction.
- **status**: Status of the property (e.g., for sale, sold).
- **price**: Selling price of the property.
- **bed**: Number of bedrooms.
- **bath**: Number of bathrooms.
- **acre_lot**: Size of land in acres.
- **street**: Street information of the property.
- **city**: City where the property is located.
- **state**: State where the property is located.
- **zip_code**: Postal code of the property.
- **house_size**: Size of the house in square feet.
- **prev_sold_date**: Date when the property was previously sold.

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2. Tools Used and Components

- **Alteryx Components**
  - We used Alteryx for data preprocessing, cleaning, and transformation:
    - **Input Data Tool**
      - Used to load the dataset into Alteryx.

- **Record ID Tool**
  - Added a unique identifier for each record to support dataset splitting.
- **Sample Tool (50%)**
  - Used to split the dataset into two equal halves.
- **Sort Tool (Descending)**
  - Used to reverse dataset order before splitting.
- **Select Tool**
  - Used for renaming columns and selecting required fields only.
- **Data Cleansing Tool**
  - Removed null values and cleaned missing or invalid data.
- **Summarize Tool**
  - Calculated:
    - Mean of price, lot size, and house size
    - Mode of bedrooms and bathrooms
- **Formula Tool**
  - Replaced null values using calculated values
  - Converted text formatting (e.g., uppercase city, state, status)
  - Split prev_sold_date into year, month, day
- **Filter Tool**
  - Removed:
    - Null/empty street values
    - Invalid sales dates
    - Properties with price > 10,000,000 (outlier filtering)
- **Output Data Tool**
  - Exported processed data into SQL Server tables.

- **SSIS Components**
  - We used SSIS for the second half of the dataset and loading into the database:
    - **Excel / Input Source**
      - Imported dataset into SSIS data flow.
    - **Derived Column**
      - Used to:
        - Handle missing values
        - Transform data types
        - Create new calculated fields
    - **Data Conversion**
      - Ensured compatibility between Unicode and non-Unicode columns (e.g., Zip Code).
    - **Conditional Split**
      - Used for filtering invalid records and separating data logic.
    - **Lookup (if applied in workflow)**
      - Used to validate or match reference data.
    - **OLE DB Destination**
      - Loaded transformed data into SQL Server tables.

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3. Dataset Splitting Process (Alteryx Logic)

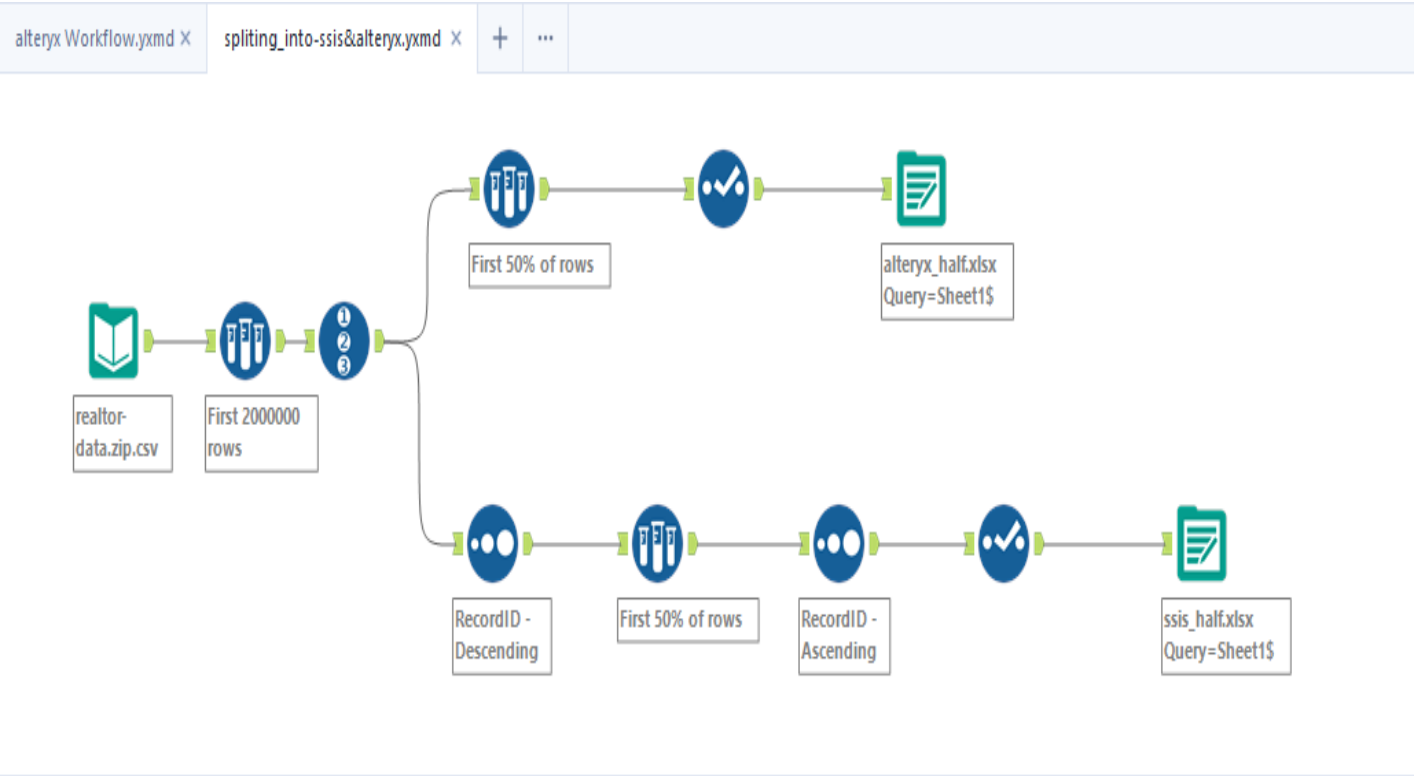
- To handle large dataset efficiently, we split it into two parts:

5. **Input Tool** → Load dataset
5. **Select Tool** → Select first 2000000 from dataset
5. **Record ID Tool** → Assign unique ID
5. **Sample Tool (50%)** → Take first half
5. **Sort Tool (Descending)** → Reverse order

5. **Sample Tool (50%) again** → Extract second half

5. **Result:**

    - Part 1 → Processed in Alteryx
    - Part 2 → Processed in SSIS



4. Data Transformations & Business Logic (Alteryx Workflow)

- **Data Input & Preparation**

- **Input Data Tool**
    - Used to import the main USA Real Estate dataset into Alteryx.
- **Select Tool**
    - Used to select only the required columns for analysis.

- Also used to rename columns for better readability and consistency.

Options Search

	Column	Type	Size	Rename	Description
>	brokered_by	Float	4	Broker_ID	
	status	V_WString	214...	Property_Status	
	price	Float	4	Price	
	bed	Int32	4	Bedrooms	
	bath	Int32	4	Bathrooms	
	acre_lot	Float	4	Lot_Size_Acres	
	street	V_WString	214...	Street_Number	
	city	V_WString	214...	City	
	state	V_WString	214...	State	
	zip_code	V_WString	214...	Zip_Code	
	house_size	Float	4	House_Size_Sqft	
	prev_sold_date	V_WString	214...	Sale_Date	
	*Unknown	Unknown	0		Dynamic or Unknow...

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5. Data Cleaning

- **Data Cleansing Tool**
  - Removed null values and missing data from the dataset.
  - Ensured data consistency by eliminating incomplete or invalid records.
- **Filter Tool**
  - Removed records containing:
    - Null or empty values in the *street* field
    - Invalid or missing values in the *sales date*

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6. Feature Engineering & Data Transformation

- **Formula Tool**

- Converted text fields to uppercase for standardization:
  - City
  - State
  - Property Status
- Split prev_sold_date into:
  - Year
  - Month
  - Day
- Replaced null values with appropriate default or calculated values.

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7. Data Aggregation (Summarization)

- **Summarize Tool**
  - Calculated the **mean (average)** for:
    - Price
    - Lot Acre
    - House Size
  - Calculated the **mode (most frequent value)** for:
    - Number of Bedrooms
    - Number of Bathrooms

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7. Outlier Handling(filtering)

- **Filter Tool**
  - Removed properties with price greater than **1000,000**
  - This step was applied to eliminate extreme outliers and improve data reliability.

8. Data Splitting & Processing Logic

- **Record ID Tool**
  - Assigned a unique ID to each record.
  - Used as a reference for dataset splitting.
- **Sample Tool (50%) + Sort Tool**
  - The dataset was split into two equal parts:
    - One processed in **Alteryx**
    - The other processed in **SSIS**
  - Sorting (descending) was applied before sampling to ensure proper distribution.

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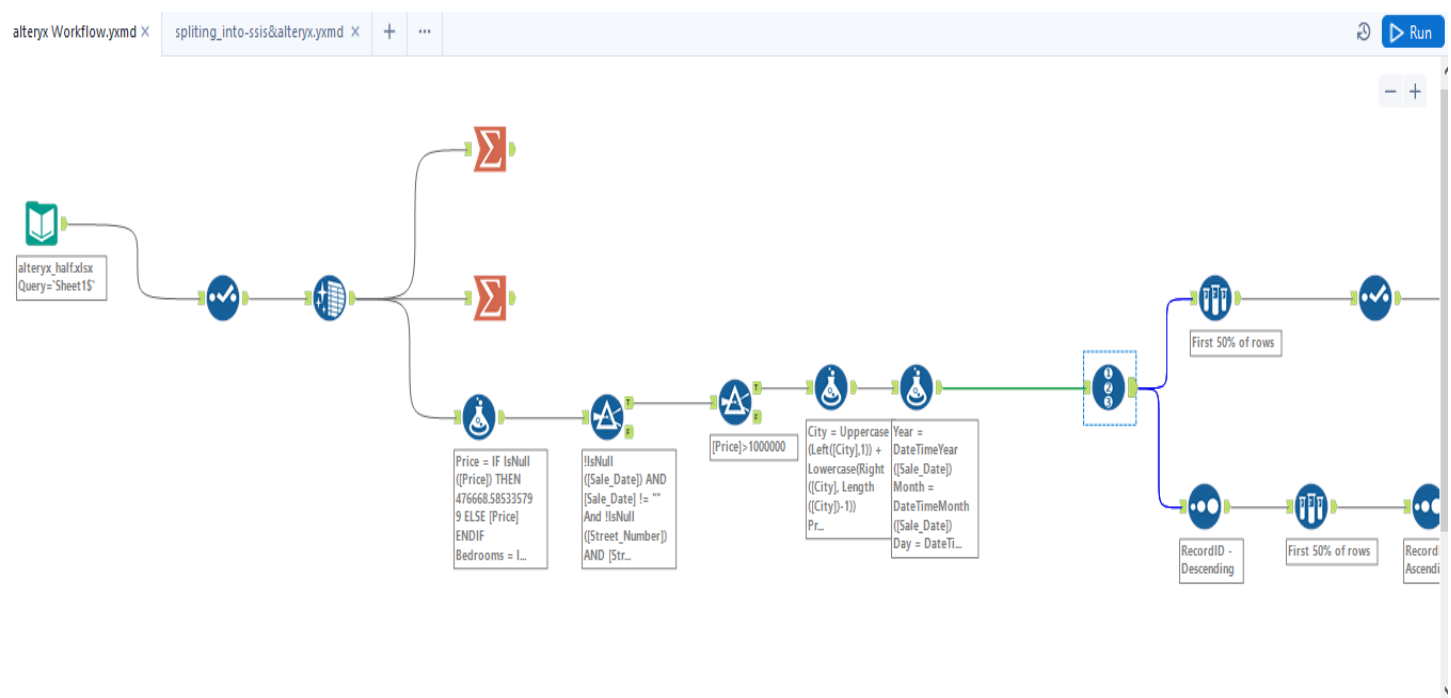
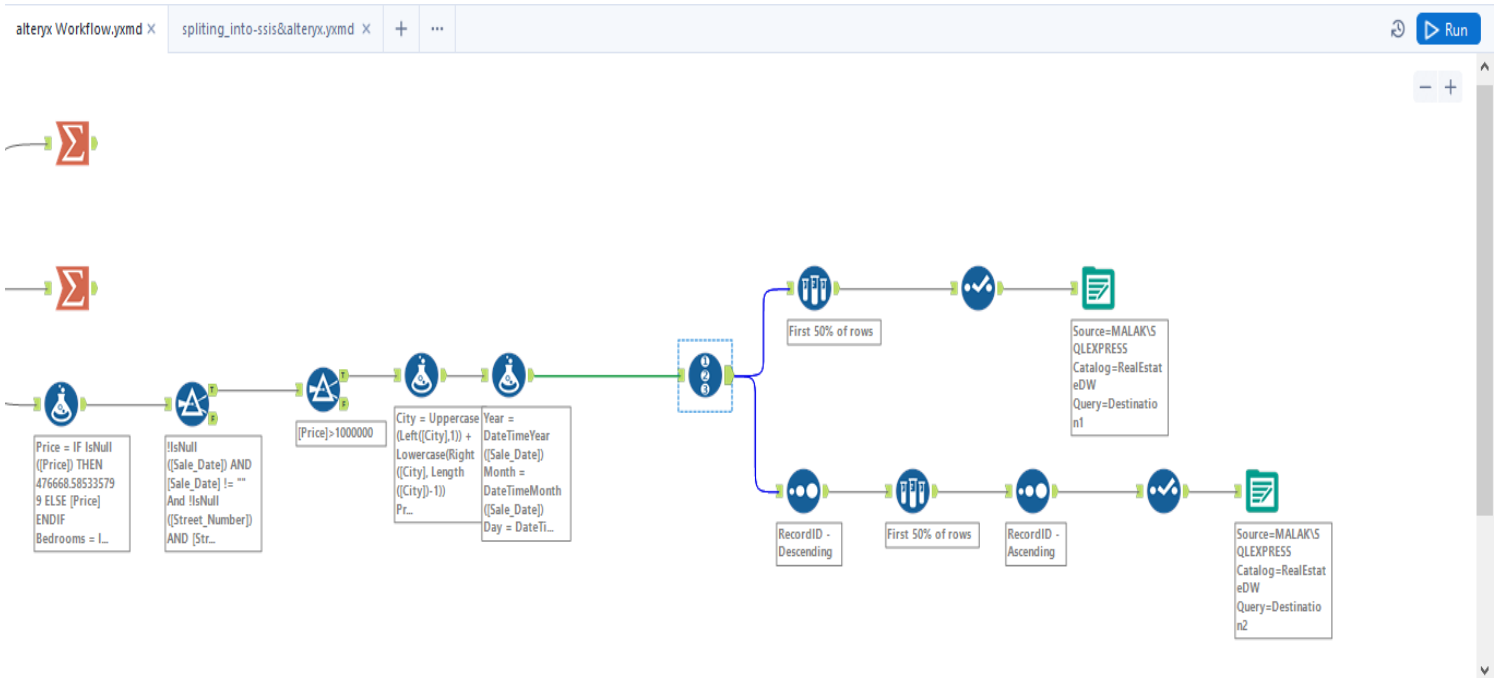
9. Data Output

- **Output Data Tool**
  - Final cleaned and transformed data was exported to a **SQL Server database**.
  - Data was stored in separate tables based on the splitting logic.





- **Alteryx part 1 is 19100 record**

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10. SSIS Data Flow Pipeline Design

- The SSIS package was designed to extract, transform, and load real estate data from an Excel source into a SQL Server database table. The data flow consists of several transformation steps to ensure data quality, consistency, and compatibility with the destination schema.

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1. Excel Source

- The process starts with an **Excel Source**, which is used as the input data provider. This source contains raw real estate data including property details such as price, bedrooms, bathrooms, lot size, house size, location, and sale date.

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2. Conditional Split (Initial Filtering)

- A **Conditional Split Transformation** is used to separate and validate specific fields such as:
  - Street Number

    - Sale Date
- This step ensures that records containing invalid or missing critical values are handled appropriately before further processing.

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3. Derived Column Transformation (Handling Null Values)

- A **Derived Column Transformation** is applied to clean missing data by replacing NULL values in the following fields:
  - Price
  - Bedrooms
  - Bathrooms
  - Lot Size (Acres)
  - House Size (Sqft)
- Default values are assigned to ensure data completeness and avoid loading errors in the database.

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4. Conditional Split (Outlier Filtering)

- Another **Conditional Split** is used to filter out high-value properties:
  - Only records where **Price > 1,000,000** are accepted for further processing.
  - This helps in removing extreme outliers and improving data quality for analysis.

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5. Derived Column Transformation (Feature Engineering)

- This transformation is used to enhance and structure the dataset by:
  - Splitting **Sale Date** into:
    - Year
    - Month
    - Day
  - Converting text fields such as **Property Status** and **Status** to **Upper Case** for standardization.

6. Derived Column Transformation (Data Formatting)

- The **City** column is cleaned by replacing spaces with underscores ("_") to ensure consistency and avoid issues during storage or querying.

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7. Percentage Sampling (Data Splitting)

- A **Percentage Sampling Transformation** is used to divide the dataset into two subsets:
  - Training / Analysis Dataset (50%)
  - Testing / Validation Dataset (50%)
- This allows for data analysis and model evaluation purposes.

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8. Data Conversion Transformation

- A **Data Conversion Transformation** is applied to ensure that all data types match the destination SQL Server table schema.
  - This step includes converting:
    - Strings to numeric types
    - Strings to date formats
    - Floating values to appropriate precision types

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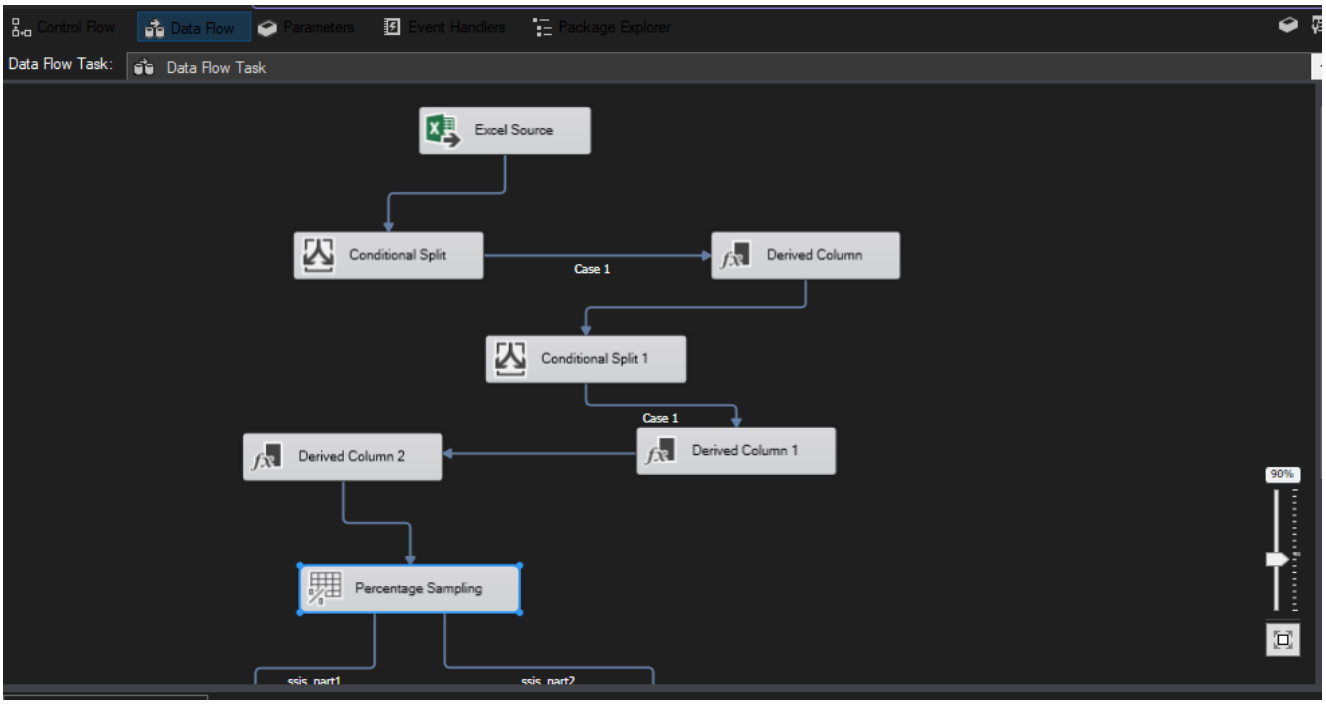
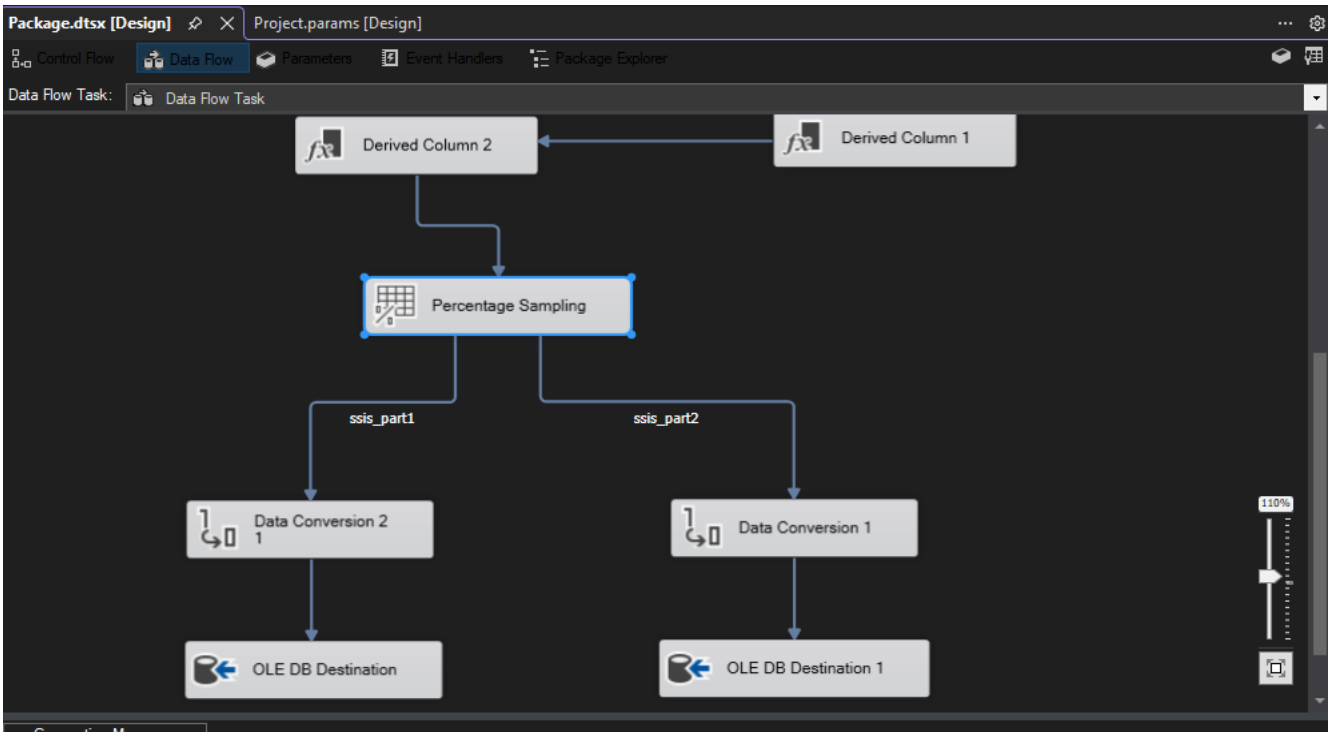
9. OLE DB Destination

- Finally, the cleaned and transformed data is loaded into the SQL Server database using an **OLE DB Destination**.
- This represents the final output layer where structured and validated data is stored for further analysis and reporting.

11. Summary

- The SSIS package ensures a complete ETL process starting from raw Excel data, performing multiple cleaning and transformation steps, and finally loading structured data into a relational database with high data quality and consistency.





Star Schema

- In order to support analytical processing and business intelligence reporting, the original property sales dataset was transformed into a Star Schema model. This structure improves query performance and simplifies reporting and visualization.
- For this project, the original flat tables Destination_1 & Destination_2 contained all descriptive and transactional attributes in one place, including property details, location details, date information, and sales price. To convert this data into a dimensional warehouse model, the dataset was divided into one fact table and two dimension tables according to the project requirements.

- **Fact Table:** Fact_Property_Sales
  - The fact table stores the measurable business value and foreign keys that reference the dimension tables. In this project, the fact table contains:
    - **Sale_ID**
    - **Property_ID**
    - **Date_ID**
    - **Price**
  - The Price column represents the main quantitative measure used for analysis, while Property_ID and Date_ID act as foreign keys.

- **Dimension Table 1:** Dim_Property
  - This dimension table stores all descriptive information related to the property itself and its location. It contains:
    - **Property_ID**
    - **Property_Status**
    - **Bedrooms**
    - **Bathrooms**
    - **Lot_Size_Acres**
    - **House_Size_Sqft**
    - **Street_Number**
    - **City**

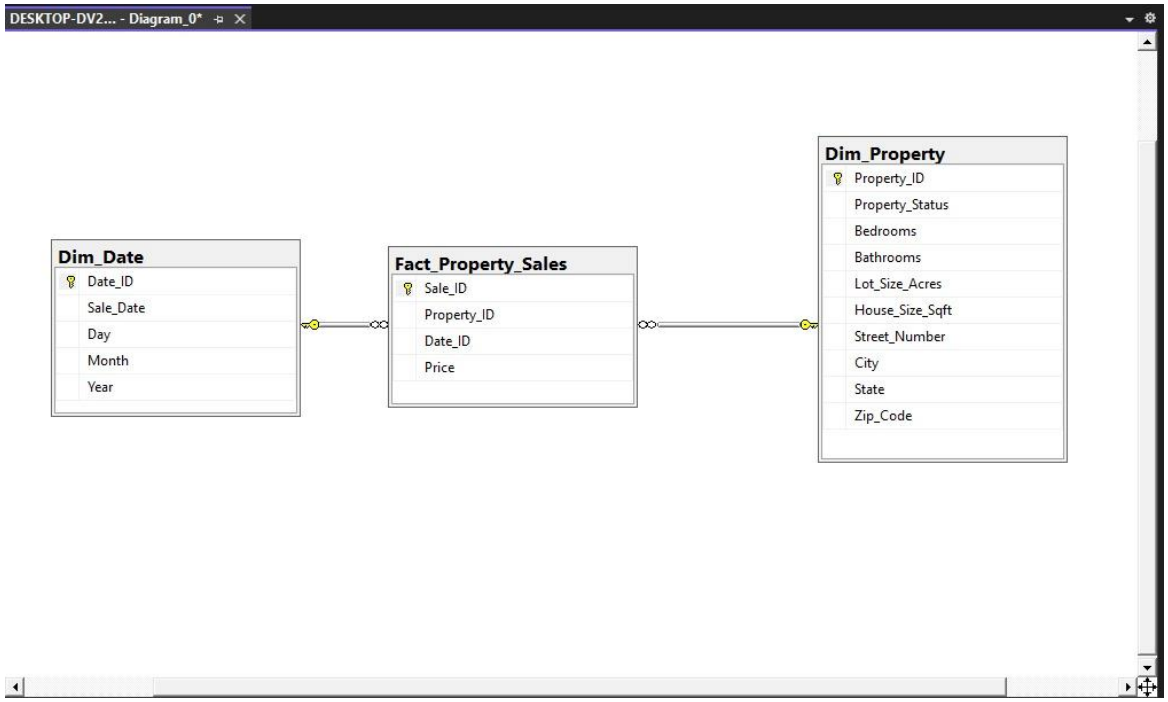
- **State**
- **Zip_Code**

  - These attributes are used for grouping, filtering, and comparing property sales.

- **Dimension Table 2:** Dim_Date
  - This dimension table stores all time-related attributes required for trend analysis. It contains:
    - **Date_ID**
    - **Sale_Date**
    - **Day**
    - **Month**
    - **Year**
  - The date dimension enables monthly and yearly sales tracking.

Star Schema Relationships

- The Fact_Property_Sales table is linked to:
  - **Dim_Property through Property_ID**
  - **Dim_Date through Date_ID**
- Thus, the final warehouse follows a valid star schema in which the fact table is located at the center and the dimension tables surround it.



Power BI Dashboard and Data Analysis

- After building the Star Schema in SQL Server, the warehouse tables were imported into Microsoft Power BI Desktop in order to perform business intelligence analysis and generate interactive visual reports.
- The imported tables were:
  - **Fact_Property_Sales**
  - **Dim_Property**
  - **Dim_Date**
- Relationships were created between the fact table and the dimension tables using Property_ID and Date_ID, forming a complete dimensional model inside Power BI.
- To enhance analytical capabilities, several DAX (Data Analysis Expressions) measures were created. These measures allow dynamic calculations based on user-selected filters and chart interactions.

DAX Measures Used

1. Total Sales

- This measure calculates the total revenue generated from all property sales.
  - **Total Sales = SUM(Fact_Property_Sales[Price])**

2. Average Price

- This measure calculates the average selling price of properties.
  - **Average Price = AVERAGE(Fact_Property_Sales[Price])**

3. Property Count

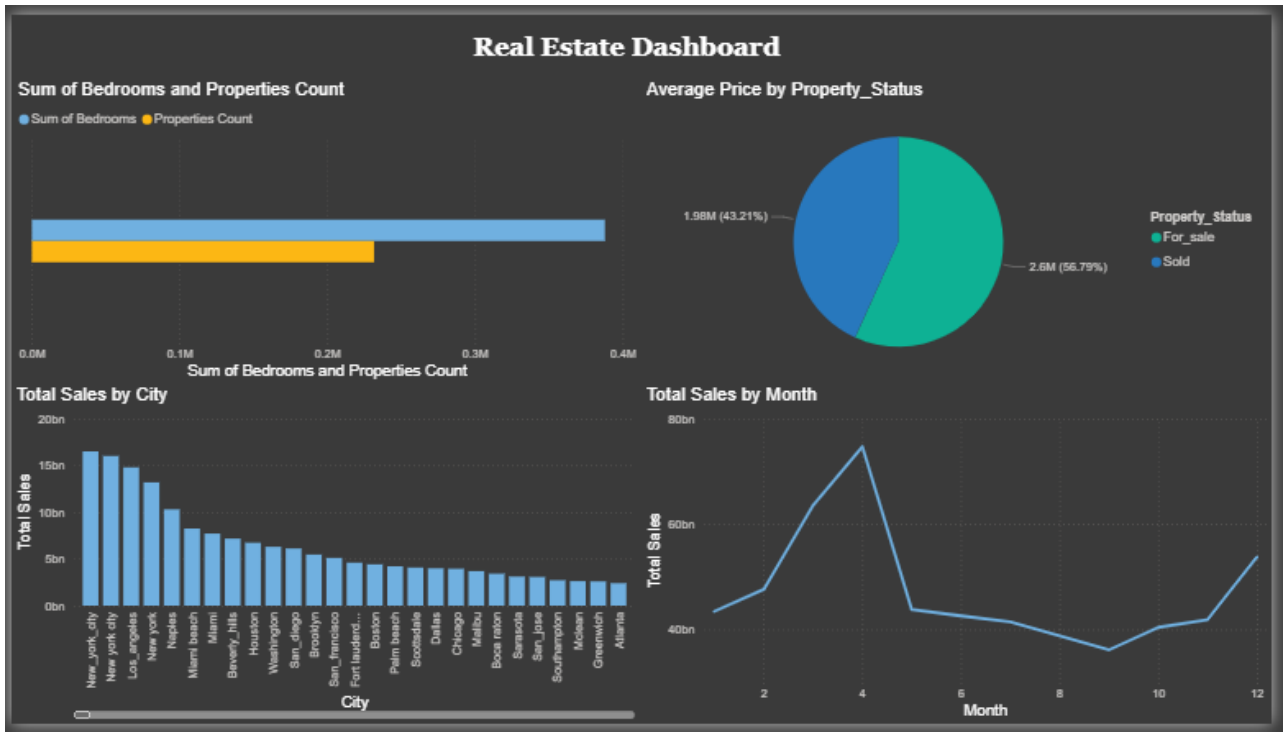
- This measure counts the total number of property transactions.
  - **Property Count = COUNTROWS(Fact_Property_Sales)**

4. Max Sale

- This measure returns the highest property sale value.
  - **Max Sale = MAX(Fact_Property_Sales[Price])**

5. **Sold Revenue**

- This measure calculates the total revenue generated only from sold properties using the CALCULATE function.
  - **Sold Revenue = CALCULATE(SUM(Fact_Property_Sales[Price]), Dim_Property[Property_Status] = "Sold")**

Visualizations Created in Power BI

- four charts were developed from the warehouse model:
  - **Total Sales by City:** compares total sales revenue among different cities.
  - **Average Price by Property Status:** compares average prices based on whether the property is sold, available, or pending.
  - **Monthly Sales Trend:** shows how property sales change over different months.
  - **Property Count by Bedrooms:** displays the number of properties available for each bedroom category.
- In addition, KPI cards were added to display summary indicators such as total sales, average price, and maximum sale value.
- The use of Power BI with DAX functions provided clear visual insights into the dataset and demonstrated the effectiveness of the designed Star Schema for analytical reporting.